

Ananda Development: Using ML- powered Google Cloud APIs to build buyer trust

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Using Cloud Speech API and Cloud Vision API, Ananda Development created a property application for mobile devices to automate condominium inspections.



About Ananda Development

Founded in 1999 and headquartered in Thailand, Ananda Development is an end-to-end property developer specializing in residential high-rise condominiums. The business has subsidiaries in construction, property management, and agency sales.

Industries: Real Estate & Construction

Google Cloud Results

- Saves approximately US\$120,000 in manpower costs over three months
- Captures operational insights and converts them into valuable data
- Builds credibility of digital transformation within the business

Location: Thailand

Save
hours
inspected
times
months

Cloud Speech to Text
(<https://cloud.google.com/speech/>)

Cloud Vision API (<https://cloud.google.com/vision/>)

The cloud, new technologies, and digital business models present rich opportunities for property developers to improve customer experiences. Thailand-headquartered Ananda Development is one property developer that recognizes the potential of these trends to transform its business.

Founded in 1999, [Ananda Development](https://www.ananda.co.th) (<https://www.ananda.co.th>) specializes in high-rise residential condominiums located within the central Bangkok metropolitan area. A prominent advocate

of innovation and modern property projects in Thailand, including smart developments that integrate digital and information technologies into residents' lives, the business is starting to use digital technologies to improve the experience of workers, buyers, and other stakeholders.

Handover, a key step

As a property developer, one of Ananda Development's most important activities is handing over a completed condominium to a buyer. The business only realizes revenue when the buyer signs the acceptance form for the property. Before this step is completed, an Ananda Development inspector and the buyer conduct detailed checks of the condominium for any build variations that need to be rectified.

"The inspection process is key to our business as it starts the process of establishing trust," explains Gao Li, Senior Manager, Business Optimization, Ananda Development. "With trust, we can unleash a world of possibilities."

However, traditional inspection processes did not help build trust. Ananda Development inspectors would check visually for problems and mark up any outcomes on paper, while in many cases a prospective buyer would also take notes and photographs of the findings. On average, a single Ananda Development inspector would have to check several hundred items per day. This workload was considerable when multiplied across several inspectors and multiple projects. Furthermore, this

laborious, manual process was subject to occasional human error that meant data could be omitted or recorded incorrectly.

Technology delivers transformation

Backed by Ananda Development Chief Executive Officer, Chanond Ruangkritya, Gao, and the company's Chief Innovation Officer, Dr. Chet Yong, decided to use technology to improve the experience of inspectors, as well as builders, buyers, project managers, and quality controllers.

The business decided to develop a property transfer mobile application that would help inspectors identify, capture, and submit defects electronically in real time and on the go as they carried out inspections.

Gao and Dr. Yong realized Ananda Development could not meet the requirements of inspectors and other stakeholders by simply deploying an out-of-the-box solution. “We decided to work closely with our inspectors to understand the circumstances in which they worked,” says Gao. “Our team spent a lot of time shadowing inspectors to truly understand their pain, and look for opportunities to improve their experiences.”

A pass-by, capture, submit process

The business determined the application would need to support a simple pass-by, capture, submit

process during inspections. Inspectors would be able to verbally describe defects and critical issues to the application running on their smartphones. The application would then track and document the inspection results. The data would then be converted into formal inspection reports and trigger work orders for third-party or in-house builders, or other tradespeople, to address. The information would also be made available to stakeholders such as project managers and quality controllers.

Ananda Development started evaluating user requirements from the second half of 2017. As well as working closely with internal workers, the innovation team also sought input from external parties that would also use information gathered through the application.

“Our inspectors are surprised by how leveraging our mobile application using Cloud APIs makes recording their findings so easy.”

*—Gao Li, Senior Manager,
Business Optimisation,
Ananda Development*

Accuracy of more than 90%

In planning the application, the business realized it would need to recognize and convert to text Thai-language speech and a version of English spoken by many Thai people. Following a review of available products, Ananda Development opted to use Cloud Speech-to-Text API (<https://cloud.google.com/speech/>). “We found the product had an accuracy rate of over 90 percent in recognizing Thai speech and high-accuracy rates in recognizing Thai English,” explains Gao.

Furthermore, Ananda Development wanted to establish a pathway to use machine learning to complete condominium inspections using unmanned drones. By using Cloud Vision API (<https://cloud.google.com/vision/>), the property developer would be able to capture images of defects and classify information about each one. “Eventually, with the help of Google, we will obtain enough data to support an algorithm to capture defects,” says Gao.

Throughout the development of the application, Ananda Development’s innovation team completed a comprehensive program of user acceptance testing and education. With development, testing, and

training completed, Ananda Development went into production and deployment at the end of 2017.

“The property transfer mobile application using Cloud Speech-to-Text API and Cloud Vision API has allowed us to transform the buyer and stakeholder experience, enhancing our brand reputation.”

*—Dr. Chet Yong, Chief
Innovation Officer,
Ananda Development*

About 4,000 units on board

Within 7 months, Ananda Development was using the property transfer mobile application to onboard about 6 projects incorporating approximately 4,000 units. At that point, application had collected between 3,000 and 4,000 findings. Ananda Development is now working to improve connectivity at some locations to help ensure the application and supporting infrastructure runs effectively. “Our inspectors are surprised by how leveraging our mobile application using Cloud APIs makes recording their findings so easy,” says Gao.

The inspection process is now lean and efficient, with inspection times reduced and the outcomes more accurate. Buyers also receive copies of electronic inspection reports and updated status notes as defects are repaired.

The business describes the benefits of the project as:

- capturing operational insights and converting them into valuable actionable data;
- delivering a centralized content management platform to address the challenges of communicating to multiple groups;
- providing real-time, on-site monitoring to enable managers to make better decisions;
- analyzing construction and vendor quality to deliver continuous improvement;
- making statutory disclosures required for compliance and safety checks; and

- paving the way for machine-learning-based defect detection.

Furthermore, inspectors can now complete their activities during the working day rather than spend extra time at the office filling out reports.

“Google has enabled us to create a highly specific digital process and demonstrate success in an industry that is not natively digital.”

—Dr. Chet Yong, Chief
Innovation Officer,
Ananda Development

130 hours and US\$120,000 saved

Within 3 months of implementation, Ananda Development had saved about 130 hours of

inspection times and about 4 million Thai baht (approximately US\$120,000) in manpower costs.

“The property transfer mobile application using Cloud Speech-to-Text API and Cloud Vision API has allowed us to transform the buyer and stakeholder experience, enhancing our brand reputation,” Dr. Yong says.

“Google has enabled us to create a highly specific digital process and demonstrate success in an industry that is not natively digital,” he adds. “This early payoff allows us to build credibility and a platform from which to drive digital transformation further into our business and the industry as a whole.”